

Leksell gamma knife treatment of uveal melanoma

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Object. The purpose of this study was to analyze treatment results, radiation-induced side effects, and prognostic survival factors for patients with uveal melanoma.

Methods. Eighty-one patients with uveal melanoma were treated using the Leksell gamma knife during a period of 6 years (1996–2001). There were 45 men and 36 women with a median age of 59 years (range 22–85 years). Seventy-five of these patients underwent minimal follow up 10 months after treatment. After patient eye immobilization, magnetic resonance (MR) imaging was performed to enable stereotactic localization. A scoring system was used to measure radiation side effects. The median target volume was 640 mm³, and the median applied minimal dose was 31.4 Gy. All patients were examined by an ophthalmologist and with MR imaging at regular intervals. Factors influencing post-treatment survival and side effects were statistically analyzed.

Conclusions. Local tumor control in the 75 patients who underwent minimal follow up after 10 months was achieved in 63 patients (84%), whereas progression was observed in 12 patients (16%). The most frequent side effect was secondary glaucoma, which was detected in 18 patients (25%). The incidence of this side effect was significantly higher when the total volume of peripheral isodose was greater than 1000 mm³ ($p = 0.015$). Toxicity in the optic nerve here was also significantly higher when the maximum dose to this structure was higher than 9 Gy ($p = 0.011$), in the cornea when the maximum dose was higher than 15 Gy ($p = 0.010$), and in the lens when the maximum dose was higher than 10 Gy ($p = 0.035$). Altogether three pretreatment variables (patient age, tumor location, and dissemination of the disease) and one treatment variable (the minimum dose applied) were identified as having a significant influence on a patient's survival.

KEY WORDS • stereotaxy • radiosurgery • uveal melanoma • local tumor control • gamma knife • complication • secondary glaucoma • prognostic factor

UVEAL melanoma represents the most common malignant primary tumor of the eye in adults. Several therapeutic options are available including resection, or enucleation, brachytherapy, and stereotactic irradiation.^{1–11,17–25} Despite the range of treatment modalities, the prognosis of this disease remains serious. Its dissemination to other organs represents the cause of death in the majority of patients. Stereotactic irradiation allows the application of high-dose radiation to a small, well-defined target volume, with a reduced dose affecting the surrounding normal structures and tissues. Stereotactic irradiation entails the use of either the Leksell gamma knife or a stereotactic linear accelerator. The effect of a single GKS treatment on rabbit eye melanoma was studied, and total destruction of the tumor in this experimental model was reported after a single dose of 60 to 90 Gy.¹⁷ Subsequently, in several reports of other treatment modalities local tumor control rates between 82 to 98% were reported.^{8,9,12,13,16–19} The choice of treatment modality depends in

particular on the location of the tumor within the eye, the tumor volume, and its dissemination into other organs.

The purpose of this study was to evaluate statistically the treatment results and side effects for those patients treated for uveal melanoma in our department. Prognostic factors for survival will also be assessed.

Clinical Material and Methods

Patient Characteristics

During a period of 6 years (1996–2001), 81 patients with uveal melanoma were treated by GKS in our department. The median age of these patients was 59 years (range 22–85 years). Seventy-five underwent follow up for a minimum of 10 months after the treatment. A statistical evaluation of the treatment effects, complications, and the survival was performed.

Treatment involved immobilization of the eye, which was achieved by applying two sutures in the rectus muscles and attaching these sutures to the frame. Sutures prevented any movement of the eye during stereotactic imag-

Abbreviations used in this paper: GKS = gamma knife radiosurgery; MR = magnetic resonance.

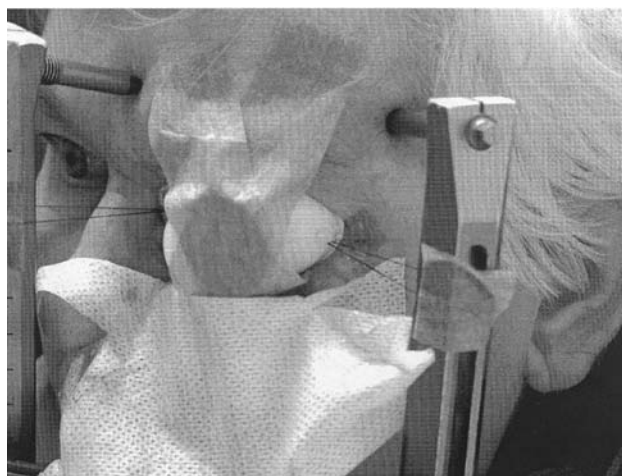


FIG. 1. Photograph. Two sutures applied to rectus muscles and attached to frame were used to prevent the movement of the eye bulb during irradiation. The eye was covered by the tissue-equivalent bolus to reduce errors in dose calculations, especially for tumors located close to the surface of the eye.

ing and irradiation. The eye was covered by a tissue-equivalent bolus to reduce errors in dose calculations, especially when tumors were located close to the surface of the eye (Fig. 1). Stereotactic MR imaging was performed for tumor localization in all patients. The Leksell GammaPlan treatment planning system was used for the calculation of the dose. The SOMA LENT scoring system was used to measure the radiation-induced side effects. Grades 3 and 4 represent serious toxicity¹⁵ (Table 1).

The following ophthalmological methods were used for diagnostic investigation and regular patient follow up after GKS: ultrasonography, indocyanin and fluorescein angiography, optical coherence tomography, ultrabiomicroscopy (for locations anterior to equator), visual function, and biomicroscopy.

The location of tumor was anterior to the equator in nine patients (12%), posterior to the equator in 47 patients (63%), and on both sides of the equator or involving the whole eye in 19 patients (25%).

The median tumor margin dose was 31.4 Gy (range 20–76.5 Gy). The median target volume was 640 mm³ (range 32–7800 mm³). The median tumor height was 8.5 mm (range 4–15 mm). Radiosurgery in 40 patients involved one isocenter. In the remaining 41 cases in which the lesions were irregular a median of three isocenters (range two–nine isocenters) were used. In 38 patients var-



FIG. 2. Left: Uveal melanoma before GKS. Right: Tumor regression observed 18 months after GKS.

ious plugging patterns were used to achieve a higher level of conformity. The median follow-up period was 32 months (range 10–74 months). In 19 patients (23.5%) spread of disease to other organs had already been detected at the time of GKS. All patients were examined by an ophthalmologist at the 1st week and the 1st month after GKS, then 3 months after GKS and thereafter every 6 months. Magnetic resonance imaging was performed at 12-month intervals. Local disease control was defined as freedom from tumor progression and was measured by a combination of ultrasonography and MR imaging.

Statistical Analysis

To evaluate the complications and side effects following GKS, toxicity in the following structures was studied as a function of the maximum dose: optic nerve, cornea, and lens. Additionally, the development of secondary glaucoma after GKS was studied as a function of the prescribed margin dose, tumor localization, and total volume of the prescription isodose. To determine the statistical significance of these factors, the chi-square test with a Yates correction factor, or the Fisher exact test (when the expected value was < 5), were used. The variable was considered to be significant when the probability value was less than 0.05.

Ten different variables were proposed as potential factors influencing survival after irradiation of uveal melanoma using the GKS. These factors were divided into two groups: pretreatment and treatment. Pretreatment variables included sex, age, tumor localization, tumor volume, and dissemination of the disease. Treatment variables included the total number of isocenters, prescribed margin isodose, total volume of the prescription isodose, maximum tumor dose, and minimum tumor dose.

TABLE 1

The SOMA LENT scoring system for measuring the radiation toxicity after uveal melanoma treatment by GKS

Ocular Component	Grade 1	Grade 2	Grade 3	Grade 4
iris	rubeosis only	rubeosis, increased intraocular pressure	neovascular glaucoma w/ ability to count fingers at 1 m	neovascular glaucoma w/out ability to count fingers at 1 m, complete blindness
optic nerve	afferent pupillary defect w/ normal-appearing nerve	≤1/4 pallor w/ asymptomatic visual field defect	>1/4 pallor or central scotoma	profound optic atrophy, complete blindness
cornea	increased tearing on exam	noninfectious keratitis	infectious keratitis, corneal ulcer	panophthalmitis, corneal scar, ulceration leading to perforation/loss of globe
lens	asymmetric lenticular opacities, no visual loss	moderate lenticular changes w/ mild–moderate visual loss	moderate lenticular changes w/ severe visual loss	severe lenticular changes

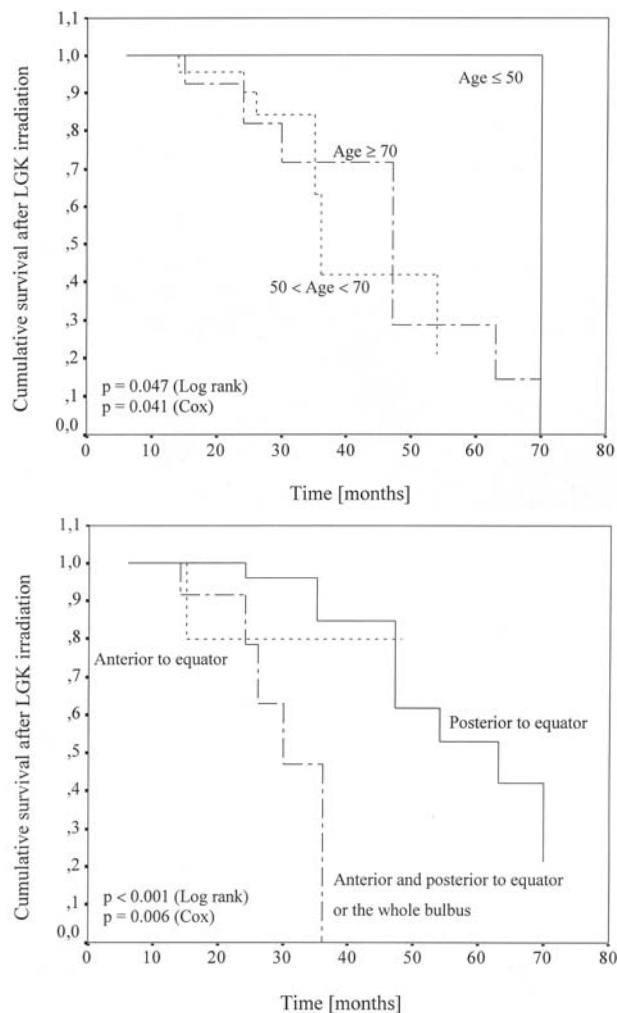


FIG. 3. Upper: Kaplan–Meier survival curves for 75 patients with uveal melanoma after GKS based on age. Lower: Kaplan–Meier survival curves for 75 patients with uveal melanoma after GKS based on tumor localization.

To determine the potential factors influencing survival, univariate and multivariate analyses were conducted. Univariate analyses were performed using Kaplan–Meier statistics with the log-rank test. Multivariate analyses were performed with the Cox proportional hazards model with the backward-stepwise (conditional likelihood ratio) method. Variables with significant probability values ($p < 0.05$) at least in one of two actuarial analyses were considered possible factors affecting survival.

All statistical analyses in this study were performed using the statistical software package SPSS version 10.

Results

Treatment Results

Fifty-seven patients are still alive after a median survival period of 15 months (range 6–70 months). Sixteen patients died (of other tumor dissemination) after a median survival period of 35 months (range 14–70 months). Local tumor control was achieved in 65 (87%) of 75 patients after a minimum follow up of 10 months. Tumor regression was observed in 45 patients (60%), stabiliza-

tion in 18 (24%), and progression in 12 (16%). Ten of the patients with tumor progression underwent enucleation. Improvement in visual function was observed in 12 patients (15%). A case in which tumor regression occurred after 18 months is illustrated in Fig. 2.

Side Effects After Irradiation

Radiation-induced late toxicity was measured and evaluated at regular intervals by ophthalmologists and MR imaging. The most frequent and severe radiation-related toxicity was secondary neovascular glaucoma in 18 (24%) of the 75 patients, and four patients underwent enucleation for this reason. The prescribed margin dose and tumor localization did not prove significant for the induction of secondary glaucoma. There was a significantly higher incidence of secondary glaucoma observed in cases in which the total volume of the prescription isodose exceeded 1000 mm³ ($p = 0.015$).

There were other types of side effects. In eight patients there was corneal damage with a median score of 3 (range 1–4). In 10 patients there was optic nerve damage, with a median score of 2 (range 1–3). In 11 patients there was damage to the iris with a median score of 2 (range 1–4). Vitreous hemorrhage was observed in four patients.

A significantly higher toxicity score (3.4) was observed in the optic nerve when the maximum dose to this structure was higher than 9 Gy ($p = 0.011$). For the cornea a maximum dose greater than 15 Gy ($p = 0.01$) was associated with a higher toxicity score. For the lens the maximum dose associated with a higher toxicity score was 10 Gy ($p = 0.035$).

Survival After Irradiation

Fifty-seven of 81 patients are still alive after a median survival period of 15 months (range 6–74 months). Sixteen patients died of tumor dissemination to other organs after a median survival period of 35 months (range 14–70 months).

Altogether three pretreatment variables (age, tumor localization, and disseminated disease) and one treatment variable (the minimal tumor dose) were identified as significantly influencing patient survival after GKS. The Kaplan–Meier survival curves are presented in Figs. 3 and 4.

Discussion

Melanoma is a malignant disease that is difficult to treat wherever it arises. It is typically resistant to standard fractionated radiotherapy and chemotherapy. Better treatment results have been observed after higher-dose fractions (a so-called hypofractionated scheme) and by immunotherapy in cases in which tumor dissemination occurred. Wide variations in the grading of malignancy can be observed, from cases of relatively benign types and several years' survival without dissemination to cases involving other tumors that have fast multiorgan dissemination, leading to death within a matter of a few months.

Radiosurgery is used to treat patients with uveal melanoma and fulfills the primary objective of protecting patients from the effects of enucleation. There are other ocular-preserving treatment modalities that provide local

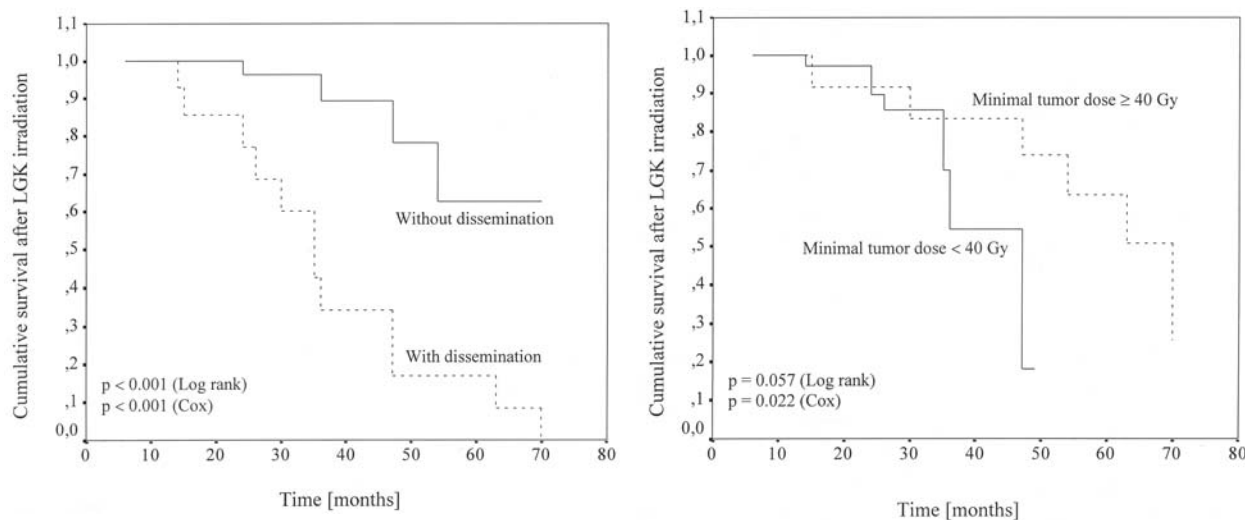


FIG. 4. *Left:* Kaplan–Meier survival curves for 75 patients with uveal melanoma after GKS based on disease dissemination. *Right:* Kaplan–Meier survival curves for 75 patients with uveal melanoma after GKS based on the minimum tumor dose.

tumor control ranging between 82 and 98%.^{1–11,15,17,19–24} Local tumor control was achieved in 65 (87%) of our patients and regression of tumor height in 45 (60%). Zehetmayer, et al.,²⁶ reported an incidence of tumor height reduction in 97% of patients. Lower applied doses and larger tumor volumes may have produced the poorer treatment results in the 75 patients presented in our study. The median applied dose was 31.4 Gy. We treated 20 patients (27%) with tumor volumes ranging from 1000 to 1500 mm³, and in 11 patients (15%) tumor volumes were greater than 1500 mm³. The median period for regression was 12 months (6–36 months). Consequently, a chance exists that regression might occur in patients after more than 12 months posttreatment.

The relatively high incidence of late complications caused by the application of a high single dose underscores the need to identify the most efficacious doses suitable for tumor volumes in GKS. This dose should ensure adequate local tumor control and an acceptable level of late toxicity. A fractionation scheme might have a sparing effect on critical structures, but because of the need for eye fixation during GKS, this would be difficult to achieve.

Differences exist in the occurrence of severe complications, which are based on radiation-induced damage to the various eye structures. The radiation tolerance of the optic nerve has been well defined in other radiosurgery procedures, in particular the irradiation of pituitary adenomas and craniopharyngiomas. The tolerance dose of the optic nerve has been recorded and ranges from 8 to 10 Gy.^{14,18} It seems more accurate to create a dose–volume histogram of the critical structure rather than a dose measurement of a single point to evaluate potential side effects. This is easy to achieve in the case of the lens and optic nerve where the whole structural volume and its margins are well defined and MR imaging can be used. On the other hand, this is not simple for other tissues such as the retina, sclera, and iris. It is also relatively easy to treat radiation-induced complications involving the lens, but atrophy of the optic nerve, corneal ulcers, and secondary glaucoma represent far more serious problems. Secondary glaucoma

is the most frequent side effect. Zehetmayer, et al.,²⁵ reported that 13% of their patients required eye enucleation as a result of this complication. In these patients requiring secondary enucleation the initial tumor height exceeded 9 mm, and ciliary body tumors were also present. The authors of a proton-beam irradiation study found that ciliary body involvement, a tumor height of greater than 8 mm, and the proximity of the tumor to the fovea were the leading risk factors for enucleation.¹⁰

The occurrence of secondary glaucoma as a typically late side effect mainly depends on the time interval of the follow up and on the use of different radiotherapy techniques. Because of this, the rate of enucleation ranges from 6% at a 2-year follow up to 19% 8 years after treatment.^{2,3,6,10,25}

In our series the incidence of secondary glaucoma was statistically higher in cases in which the volume of treatment isodose was greater than 1000 mm³. The aim of GKS is not only to preserve the eye and its visual function, but, by achieving sufficient local control, to decrease the potential for hematological dissemination. There were statistically significant longer survival periods observed when the minimum dose applied was 40 Gy and higher.

Conclusions

Stereotactic radiosurgery is a safe and minimally invasive method by which to treat uveal melanoma. In our study it provided a sufficient level of local control in 84% of patients.

Radiosurgery is an alternate treatment modality for uveal melanoma, protecting against the effects of enucleation, and in some cases preserving visual function. Improvement in vision was noticed in 12 (15%) of our patients.

The detection of factors influencing complications and the definition the dosage tolerances of critical eye structures can help to determine which types of tumors (in terms of volume, height, and location) are acceptable for radiosurgery. The most frequent complication was sec-

ondary glaucoma in 24% of patients. Other serious statistically significant complications occurred after maximum doses of 9 Gy applied to the optic nerve, 10 Gy to the lens, and 15 Gy to the cornea.

Statistically significant positive prognostic factors for survival included age (< 50 years), lesions located behind the equator, a minimum applied dose of 40 Gy, and absence of tumor dissemination to other organs.

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